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CO2 – AT1

MLA 0201 – Fundamentals of Machine Learning

Annexure A

Analytical Problem Solving

Q1.A real estate company collected the following training data:(5Marks -10 Mins)

House No	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a linear regression model of the form

$$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Q2.An email system uses the following feature values:(5Marks -10 Mins)

Email	Word Frequency (Offer)	Word Frequency (Win)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Q3.A medical dataset is given:(5Marks -10 Mins)

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Q4.Given the following dataset:(5Marks -10 Mins)

Data Point	X ₁	X ₂	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct a discriminant function:

$$g(x) = w_1x_1 + w_2x_2 + w_0$$

(ii) Determine the decision boundary separating the two classes.

(iii) Classify a new point (X₁ = 3, X₂ = 3).

Q5.A telecom company collected the following data:(5Marks -10 Mins)

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
3	400	15	0
4	900	5	1
5	600	10	1

- Formulate the Bayesian logistic regression model.
- Estimate model parameters using the dataset.
- Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Q6. Given the following dataset for customer classification: (5 Marks - 10 Mins)

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

For a new customer (Age = 32, Income = 50):

- Compute Euclidean distance from all points
- Classify using $k = 3$
- Determine the final class label

Q7. Given the dataset: (5 Marks - 10 Mins)

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

- Compute Entropy of Play

- (ii) . Calculate Information Gain for attribute Outlook
- (iii). Identify the root node of the decision tree

Q8.Given data points:(5Marks -10 Mins)

X1	X2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

- (i) .Plot the points
- (ii) .Determine a linear separating hyperplane
- (iii).Find the equation of the decision boundary

Q9.Given confusion matrices:(5Marks -10 Mins)

Logistic Regression:

	Pred +	Pred -
Actual +	40	10
Actual -	5	45

Naïve Bayes:

	Pred +	Pred -
Actual +	35	15
Actual -	10	40

- (i).Compute Accuracy, Precision, Recall for both models
- (ii).Compare results and identify the better model

Q10. Given points: (5 Marks - 10 Mins)

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids:

$C1 = (2, 4)$, $C2 = (8, 9)$

(i). Perform one iteration of K-means

(ii). Assign clusters

(iii). Compute new centroids

1. Linear Regression

Given Data

House No	Area (sq.ft) (X_1)	Bedrooms (X_2)	Price (₹ Lakhs) (Y)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a Linear Regression Model

A multiple linear regression model is used to predict the house price based on its area and the number of bedrooms. The mathematical form of the model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

where:

- Y = House Price (₹ Lakhs)
- X_1 = Area of the house (sq.ft)
- X_2 = Number of Bedrooms
- β_0 = Intercept (constant term)
- β_1 = Regression coefficient for Area
- β_2 = Regression coefficient for Bedrooms

The objective of the model is to estimate the price of a house using these two independent variables.

(ii) Estimate the Regression Coefficients

Using the given training dataset, the approximate regression coefficients are:

- $\beta_0 = 5$
- $\beta_1 = 0.04$
- $\beta_2 = 2.5$

Hence, the estimated regression equation is:

$$Y = 5 + 0.04X_1 + 2.5X_2$$

where:

- Constant (Intercept) = **5**
- Area coefficient = **0.04**
- Bedroom coefficient = **2.5**

(iii) Interpretation of the Coefficients

Intercept ($\beta_0 = 5$)

The intercept represents the estimated base price of a house when both the area and the number of bedrooms are zero. Although such a situation is not realistic, the intercept acts as the starting value of the regression equation.

Area Coefficient ($\beta_1 = 0.04$)

The coefficient **0.04** indicates that for every additional **1 square foot** increase in the house area, the predicted house price increases by approximately **0.04 lakhs (₹4,000)**, assuming the number of bedrooms remains constant.

Bedroom Coefficient ($\beta_2 = 2.5$)

The coefficient **2.5** means that adding **one extra bedroom** increases the predicted house price by approximately **₹2.5 lakhs**, while keeping the house area constant.

Final Regression Model:

$$\text{Price (₹ Lakhs)} = 5 + 0.04 \times \text{Area} + 2.5 \times \text{Bedrooms}$$

2. Linear Classification

Given Data

Email	Word Frequency (Offer) (X_1)	Word Frequency (Win) (X_2)	Spam (Y)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

Where:

- **Spam = 1** → Spam Email
- **Spam = 0** → Not Spam

(i) Construct a Linear Classification Model

A linear classification model predicts whether an email is **Spam** or **Not Spam** based on the frequency of the words "**Offer**" and "**Win**".

The general form of the linear classification model is:

$$y = w_1X_1 + w_2X_2 + b$$

where:

- **y** = Classification score
- **X₁** = Word Frequency of **Offer**
- **X₂** = Word Frequency of **Win**
- **w₁, w₂** = Weights (coefficients)
- **b** = Bias (intercept)

Using the given dataset, an approximate linear model is:

$$y = X_1 + X_2 - 2$$

This model combines the frequencies of the words **Offer** and **Win** to classify an email.

(ii) Determine the Decision Boundary

The decision boundary is obtained by setting the classification score equal to zero.

$$X_1 + X_2 - 2 = 0$$

Therefore,

$$X_1 + X_2 = 2$$

Classification Rule

- If $X_1 + X_2 > 2 \rightarrow$ **Spam (1)**
- If $X_1 + X_2 \leq 2 \rightarrow$ **Not Spam (0)**

Thus, the line $X_1 + X_2 = 2$ separates the spam emails from the non-spam emails.

(iii) Classify a New Email

Given:

- **Offer = 2**
- **Win = 1**

Substitute these values into the model:

$$y = 2 + 1 - 2$$

$$y = 1$$

Since the score is **greater than 0**, the email is classified as:

$$\text{Spam (1)}$$

Linear Classification Model:

$$y = X_1 + X_2 - 2$$

Decision Boundary:

$$X_1 + X_2 = 2$$

Prediction for New Email:

For **Offer = 2** and **Win = 1**,

$$y = 1 > 0$$

$$\text{Predicted Class} = \text{Spam (1)}$$

3. Naïve Bayes Classification

Given Data

Patient	Fever (X_1)	Headache (X_2)	Disease (Y)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

Where:

- **Disease = Yes** → Patient has the disease.
- **Disease = No** → Patient does not have the disease.

(i) Compute Prior Probabilities

The **prior probability** represents the probability of each class before considering the feature values.

Total number of patients = **5**

Number of patients with **Disease = Yes** = **3**

Number of patients with **Disease = No** = **2**

Therefore,

$$P(\text{Yes}) = \frac{3}{5} = 0.6$$

$$P(\text{No}) = \frac{2}{5} = 0.4$$

Thus, the prior probabilities are:

- **P(Disease = Yes) = 0.6**
- **P(Disease = No) = 0.4**

(ii) Compute Conditional Probabilities

The **conditional probability** is the probability of a feature occurring given a particular class.

For Disease = Yes

There are **3** patients with Disease = Yes.

- Fever = 1 appears **3** times.

$$P(\text{Fever} = 1 \mid \text{Yes}) = \frac{3}{3} = 1.0$$

- Headache = 0 appears **1** time.

$$P(\text{Headache} = 0 \mid \text{Yes}) = \frac{1}{3} = 0.33$$

For Disease = No

There are **2** patients with Disease = No.

- Fever = 1 appears **0** times.

$$P(\text{Fever} = 1 \mid \text{No}) = \frac{0}{2} = 0$$

- Headache = 0 appears **1** time.

$$P(\text{Headache} = 0 \mid \text{No}) = \frac{1}{2} = 0.5$$

Thus, the conditional probabilities are:

- **P(Fever = 1 | Yes) = 1.0**
- **P(Headache = 0 | Yes) = 0.33**
- **P(Fever = 1 | No) = 0**
- **P(Headache = 0 | No) = 0.5**

(iii) Estimate the Probability that a Patient with Fever = 1 and Headache = 0 has the Disease

Using the **Naïve Bayes theorem**,

$$\begin{aligned} P(\text{Yes} \mid \text{Fever} = 1, \text{Headache} = 0) \\ \propto P(\text{Yes}) \times P(\text{Fever} = 1 \mid \text{Yes}) \times P(\text{Headache} = 0 \mid \text{Yes}) \end{aligned}$$

Substituting the values,

$$\begin{aligned} &= 0.6 \times 1.0 \times 0.33 \\ &= 0.198 \end{aligned}$$

Similarly,

$$P(No | Fever = 1, Headache = 0) \propto 0.4 \times 0 \times 0.5 = 0$$

Since,

$$0.198 > 0$$

the patient is classified as **Disease = Yes**.

Final Answer

Prior Probabilities

$$P(Yes) = 0.6, P(No) = 0.4$$

Conditional Probabilities

$$P(Fever = 1 | Yes) = 1.0$$

$$P(Headache = 0 | Yes) = 0.33$$

$$P(Fever = 1 | No) = 0$$

$$P(Headache = 0 | No) = 0.5$$

Prediction

For a patient with **Fever = 1** and **Headache = 0**,

$$\text{Predicted Class} = \text{Disease (Yes)}$$

3. Linear Discriminant Analysis

Given Data

Data Point	X ₁	X ₂	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

Where:

- **X₁** and **X₂** are the input features.
- **C1** and **C2** are the two classes.

(i) Construct a Discriminant Function

A **Linear Discriminant Analysis (LDA)** model is used to classify data points into different classes by finding a linear function that best separates the classes.

The general form of the discriminant function is:

$$g(X) = w_1X_1 + w_2X_2 + b$$

where:

- **g(X)** = Discriminant score
- **X₁** = First feature
- **X₂** = Second feature
- **w₁, w₂** = Weights (coefficients)
- **b** = Bias (intercept)

Using the given dataset, an approximate discriminant function is:

$$g(X) = X_1 + X_2 - 8$$

This function is used to distinguish between **Class C1** and **Class C2**.

(ii) Determine the Decision Boundary

The **decision boundary** is obtained by setting the discriminant function equal to zero.

$$X_1 + X_2 - 8 = 0$$

Therefore,

$$X_1 + X_2 = 8$$

Classification Rule

- If $X_1 + X_2 < 8 \rightarrow$ **Class C1**
- If $X_1 + X_2 \geq 8 \rightarrow$ **Class C2**

Thus, the line $X_1 + X_2 = 8$ separates the two classes.

(iii) Classify a New Point

Given:

- $X_1 = 3$
- $X_2 = 3$

Substitute these values into the discriminant function:

$$g(X) = 3 + 3 - 8$$

$$g(X) = -2$$

Since the value is **less than 0**, the point belongs to:

$$\text{Class C1}$$

Final Answer

Discriminant Function

$$g(X) = X_1 + X_2 - 8$$

Decision Boundary

$$X_1 + X_2 = 8$$

Prediction

For the new point (3,3),

$$g(X) = -2$$

$$\text{Predicted Class} = \text{C1}$$

5. Bayesian Logistic Regression

Given Data

Customer	Monthly Charges (X ₁)	Tenure (Months) (X ₂)	Churn (Y)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

Where:

- **X₁** = Monthly Charges
- **X₂** = Customer Tenure (Months)
- **Y = 1** → Customer will Churn
- **Y = 0** → Customer will Not Churn

(i) Formulate the Bayesian Logistic Regression Model

Bayesian Logistic Regression is used to predict the probability that a customer will **churn (leave the service)** based on monthly charges and tenure.

The general logistic regression model is:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}}$$

where:

- **P(Y = 1)** = Probability of customer churn
- **β₀** = Intercept (bias)
- **β₁** = Coefficient for Monthly Charges
- **β₂** = Coefficient for Tenure
- **e** = Exponential constant

(ii) Estimate the Model Parameters

Using the given dataset, the approximate parameter values are:

- **β₀ = -4**
- **β₁ = 0.008**

- $\beta_2 = -0.25$

Hence, the estimated logistic regression model is:

$$P(Y = 1) = \frac{1}{1 + e^{-(-4 + 0.008X_1 - 0.25X_2)}}$$

Interpretation of the coefficients:

- $\beta_1 = 0.008$: As the monthly charges increase, the probability of churn also increases.
- $\beta_2 = -0.25$: As customer tenure increases, the probability of churn decreases.

(iii) Predict Whether a Customer Will Churn

Given:

- **Monthly Charges = 650**
- **Tenure = 9 months**

Substitute these values into the model:

$$z = -4 + (0.008 \times 650) - (0.25 \times 9)$$

$$z = -4 + 5.2 - 2.25$$

$$z = -1.05$$

Now compute the probability:

$$P(Y = 1) = \frac{1}{1 + e^{1.05}}$$

$$P(Y = 1) \approx 0.26$$

$$\text{Predicted Class} = \text{Churn (1)}$$

Final Answer

Bayesian Logistic Regression Model

$$P(Y = 1) = \frac{1}{1 + e^{-(-4 + 0.008X_1 - 0.25X_2)}}$$

Estimated Parameters

- $\beta_0 = -4$
- $\beta_1 = 0.008$
- $\beta_2 = -0.25$

Prediction

For **Monthly Charges = 650** and **Tenure = 9 months**,

$$\text{Predicted Class} = \text{Churn (1)}$$

6. K-Nearest Neighbours

Given Data

Customer	Age (Years)	Income (₹ Thousands)	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

New Customer: (Age = 32, Income = 50)

(i) Compute Euclidean Distance from All Points

The Euclidean distance formula is:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Distance from C1 (25, 40)

$$\begin{aligned}d &= \sqrt{(32 - 25)^2 + (50 - 40)^2} \\&= \sqrt{7^2 + 10^2} = \sqrt{49 + 100} = \sqrt{149} = 12.21\end{aligned}$$

Distance from C2 (30, 60)

$$\begin{aligned}d &= \sqrt{(32 - 30)^2 + (50 - 60)^2} \\&= \sqrt{2^2 + (-10)^2} = \sqrt{4 + 100} = \sqrt{104} = 10.20\end{aligned}$$

Distance from C3 (35, 80)

$$\begin{aligned}d &= \sqrt{(32 - 35)^2 + (50 - 80)^2} \\&= \sqrt{(-3)^2 + (-30)^2} = \sqrt{9 + 900} = \sqrt{909} = 30.15\end{aligned}$$

Distance from C4 (40, 20)

$$\begin{aligned}d &= \sqrt{(32 - 40)^2 + (50 - 20)^2} \\&= \sqrt{(-8)^2 + 30^2} = \sqrt{64 + 900} = \sqrt{964} = 31.05\end{aligned}$$

(ii) Classify Using $k = 3$

Arrange the distances in ascending order:

Customer Distance Class

C2	10.20	A
C1	12.21	A
C3	30.15	B
C4	31.05	B

The **three nearest neighbors** ($k = 3$) are:

- **C2 → Class A**
- **C1 → Class A**
- **C3 → Class B**

Votes:

- **Class A = 2**
- **Class B = 1**

(iii) Determine the Final Class Label

Since **Class A** has the majority of votes among the three nearest neighbors, the new customer is classified as:

Final Class = A

Euclidean Distances

- **Distance to C1 = 12.21**
- **Distance to C2 = 10.20**
- **Distance to C3 = 30.15**
- **Distance to C4 = 31.05**

Nearest Neighbors ($k = 3$)

- C2 (A)
- C1 (A)
- C3 (B)

Final Prediction

Predicted Class = A

7. Decision Tree using Information

Given Data

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

Where:

- **Play = Yes** → Play the game.
- **Play = No** → Do not play the game.

(i) Compute the Entropy of Play

Entropy measures the impurity or uncertainty in the dataset.

The formula for entropy is:

$$\text{Entropy}(S) = -P(\text{Yes})\log_2 P(\text{Yes}) - P(\text{No})\log_2 P(\text{No})$$

From the dataset:

- Number of **Yes** = 2
- Number of **No** = 2
- Total records = 4

Therefore,

$$P(\text{Yes}) = \frac{2}{4} = 0.5$$

$$P(\text{No}) = \frac{2}{4} = 0.5$$

Substituting into the formula,

$$\begin{aligned}\text{Entropy}(S) &= -(0.5)\log_2(0.5) - (0.5)\log_2(0.5) \\ &= -(0.5)(-1) - (0.5)(-1) \\ &= 0.5 + 0.5 = 1.0\end{aligned}$$

Hence,

$$\text{Entropy}(\text{Play}) = 1.0$$

(ii) Calculate Information Gain for Attribute Outlook

The formula for Information Gain is:

$$IG(S, A) = Entropy(S) - \text{Weighted Entropy}(A)$$

For **Outlook**:

- **Sunny** $\rightarrow$ (No, No) $\rightarrow$ Entropy = 0
- **Overcast** $\rightarrow$ (Yes) $\rightarrow$ Entropy = 0
- **Rain** $\rightarrow$ (Yes) $\rightarrow$ Entropy = 0

Since every subset is pure,

$$\text{Weighted Entropy} = 0$$

Therefore,

$$IG(Outlook) = 1.0 - 0$$

$$\text{Information Gain}(Outlook) = 1.0$$

(iii) Identify the Root Node of the Decision Tree

The attribute with the **highest Information Gain** becomes the **root node** of the decision tree.

Since,

$$IG(Outlook) = 1.0$$

which is the highest, the root node is:

$$\text{Root Node} = \text{Outlook}$$

Final Answer

Entropy of Play

$$Entropy(Play) = 1.0$$

Information Gain of Outlook

$$IG(Outlook) = 1.0$$

Root Node

$$\text{Outlook}$$

8. Support Vector Machine

Given Data

X_1	X_2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

Where:

- $+1 \rightarrow$ Positive Class
- $-1 \rightarrow$ Negative Class

(i) Plot the Points

The given data points can be represented on a 2D graph as follows:

Class Data Points

$+1$ (2,2), (4,4)

-1 (2,4), (4,2)

The positive class points lie on one diagonal, while the negative class points lie on the opposite diagonal.

(ii) Determine a Linear Separating Hyperplane

A **Support Vector Machine (SVM)** finds the best linear hyperplane that separates the two classes.

The general equation of a hyperplane is:

$$w_1X_1 + w_2X_2 + b = 0$$

For the given dataset, a suitable separating hyperplane is:

$$X_1 - X_2 = 0$$

This hyperplane divides the positive and negative class points into two separate regions.

(iii) Find the Equation of the Decision Boundary

The decision boundary is obtained from the separating hyperplane.

Thus, the equation of the decision boundary is:

$$X_1 - X_2 = 0$$

or equivalently,

$$X_1 = X_2$$

This line separates the two classes in the feature space.

Final Answer

Plot of the Points

- **Positive Class (+1):** (2,2), (4,4)
- **Negative Class (-1):** (2,4), (4,2)

Linear Separating Hyperplane

$$X_1 - X_2 = 0$$

Decision Boundary

$$X_1 = X_2$$

9. Model Evaluation using Confusion Matrix

Given Confusion Matrices

Logistic Regression

	Predicted +	Predicted -
Actual +	40	10
Actual -	5	45

Naïve Bayes

	Predicted +	Predicted -
Actual +	35	15
Actual -	10	40

The evaluation metrics are calculated using the following formulas:

- **Accuracy**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**

$$\text{Recall} = \frac{TP}{TP + FN}$$

where:

- **TP** = True Positive
- **TN** = True Negative
- **FP** = False Positive
- **FN** = False Negative

(i) Compute Accuracy, Precision, and Recall

A) Logistic Regression

From the confusion matrix:

- TP = 40
- TN = 45
- FP = 5
- FN = 10

Accuracy

$$\frac{40 + 45}{40 + 45 + 5 + 10} = \frac{85}{100} = 0.85$$

Accuracy = 85%

Precision

$$\frac{40}{40 + 5} = \frac{40}{45} = 0.8889$$

Precision = 88.89%

Recall

$$\frac{40}{40 + 10} = \frac{40}{50} = 0.80$$

Recall = 80%

B) Naïve Bayes

From the confusion matrix:

- TP = 35
- TN = 40
- FP = 10
- FN = 15

Accuracy

$$\frac{35 + 40}{35 + 40 + 10 + 15} = \frac{75}{100} = 0.75$$

Accuracy = 75%

Precision

$$\frac{35}{35 + 10} = \frac{35}{45} = 0.7778$$

$$\boxed{\text{Precision} = 77.78\%}$$

Recall

$$\frac{35}{35 + 15} = \frac{35}{50} = 0.70$$

$$\boxed{\text{Recall} = 70\%}$$

(ii) Compare the Results and Identify the Better Model

The calculated performance metrics are:

Metric	Logistic Regression	Naïve Bayes
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Accuracy	85%	75%
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Precision	88.89%	77.78%
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Recall	80%	70%
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From the comparison:

- Logistic Regression has **higher Accuracy (85%)**.
- Logistic Regression has **higher Precision (88.89%)**.
- Logistic Regression has **higher Recall (80%)**.

Therefore, **Logistic Regression performs better than Naïve Bayes** for the given dataset.

Final Answer

Logistic Regression

- Accuracy = 85%
- Precision = 88.89%
- Recall = 80%

Naïve Bayes

- Accuracy = 75%
- Precision = 77.78%
- Recall = 70%

Better Model

Logistic Regression

10. K-Means Clustering

Given Data Points

Data Point	X ₁	X ₂
P1	2	4
P2	3	5
P3	8	9
P4	9	10

Initial Centroids

$$C_1 = (2, 4)$$

$$C_2 = (8, 9)$$

K-Means clustering groups similar data points by calculating the distance between each point and the centroids.

The Euclidean distance formula is:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

(i) Perform One Iteration of K-Means

Distance Calculation

Point P1 (2,4)

Distance from C1:

$$\begin{aligned} d &= \sqrt{(2 - 2)^2 + (4 - 4)^2} \\ &= 0 \end{aligned}$$

Distance from C2:

$$\begin{aligned} d &= \sqrt{(2 - 8)^2 + (4 - 9)^2} \\ &= \sqrt{36 + 25} \\ &= \sqrt{61} = 7.81 \end{aligned}$$

P1 is assigned to **C1**.

Point P2 (3,5)

Distance from C1:

$$\begin{aligned}d &= \sqrt{(3-2)^2 + (5-4)^2} \\&= \sqrt{1+1} \\&= 1.41\end{aligned}$$

Distance from C2:

$$\begin{aligned}d &= \sqrt{(3-8)^2 + (5-9)^2} \\&= \sqrt{25+16} \\&= \sqrt{41} = 6.40\end{aligned}$$

P2 is assigned to **C1**.

Point P3 (8,9)

Distance from C1:

$$\begin{aligned}d &= \sqrt{(8-2)^2 + (9-4)^2} \\&= \sqrt{36+25} \\&= 7.81\end{aligned}$$

Distance from C2:

$$\begin{aligned}d &= \sqrt{(8-8)^2 + (9-9)^2} \\&= 0\end{aligned}$$

P3 is assigned to **C2**.

Point P4 (9,10)

Distance from C1:

$$\begin{aligned}d &= \sqrt{(9-2)^2 + (10-4)^2} \\&= \sqrt{49+36} \\&= 9.21\end{aligned}$$

Distance from C2:

$$\begin{aligned}d &= \sqrt{(9-8)^2 + (10-9)^2} \\&= \sqrt{1+1}\end{aligned}$$

$$= 1.41$$

P4 is assigned to **C2**.

(ii) Assign Clusters

After comparing distances:

Cluster 1 (C1)

$$\boxed{P1(2, 4), P2(3, 5)}$$

Cluster 2 (C2)

$$\boxed{P3(8, 9), P4(9, 10)}$$

(iii) Compute New Centroids

The new centroid is calculated using:

$$\text{Centroid} = \left(\frac{\sum X_1}{n}, \frac{\sum X_2}{n} \right)$$

New Centroid of Cluster 1

Points:

$$(2, 4), (3, 5)$$

$$X_1 = \frac{2 + 3}{2} = 2.5$$

$$X_2 = \frac{4 + 5}{2} = 4.5$$

Therefore,

$$\boxed{C_1 = (2.5, 4.5)}$$

New Centroid of Cluster 2

Points:

$$(8, 9), (9, 10)$$

$$X_1 = \frac{8 + 9}{2} = 8.5$$

$$X_2 = \frac{9 + 10}{2} = 9.5$$

Therefore,

$$C_2 = (8.5, 9.5)$$

Final Answer

Cluster Assignment

Cluster 1:

$$(2, 4), (3, 5)$$

Cluster 2:

$$(8, 9), (9, 10)$$

New Centroids

$$C_1 = (2.5, 4.5)$$

$$C_2 = (8.5, 9.5)$$